

The one caveat here is that the manager needs to further specify the underlying execution strategy. For example, if the strategy is a VWAP strategy, then the corresponding POV rate is:

$$POV = \left(\frac{Size}{1 + Size} \right)$$

Therefore, we are not able to solve Equation 11.9 in terms of size directly. But we can determine the size via fairly straightforward numerical methods but we still need to solve for each stock individually.

The size corresponding to a cost of 20 bp using a full day VWAP strategy is found from the following equation where we substitute POV with

$$\frac{Size}{1 + Size}.$$

$$20 = b_1 \cdot (a_1 \cdot Size^{a_2} \cdot \sigma^{a_3}) \cdot \left(\frac{Size}{1 + Size} \right)^{a_4} + (1 - b_1) \cdot (a_1 \cdot Size^{a_2} \cdot \sigma^{a_3}) \quad (11.10)$$

Notice that in this formulation the portfolio manager can also incorporate stress testing to determine the position size that will result in a cost of 20 bp under adverse market conditions by changing the volatility and liquidity.

Figure 11.6b shows the holding sizes for SP500 stocks resulting in a trading cost of 20 bp (the average cost for an order of 10% ADV). The average position size is about 10% as expected and the range is from 4 to 76% ADV. (The figure truncates the scale at 35%). Figure 11.7b shows the position sizes for R2000 stocks resulting in a trading cost of 37 bp (the average cost for an order of 10% ADV). The average position size is again about 10% and the range in order sizes is from 2 to 67% ADV.

This analysis shows that there are many stocks the manager can hold in the portfolio at more than 10% ADV and not worry about incurring unnecessary transaction costs, even under stressed market conditions. And if these stocks do have appealing alpha expectations the manager can greatly enhance portfolio performance by determining appropriate order sizes to maximize profits. Additionally, there are other stocks that should not be held even at the 10% ADV level because the corresponding costs will be too high and will erode too much of the uncovered alpha. These are the stocks that need to be held in much lower quantities or possibly not held in the portfolio at all. Unfortunately, the computational process above does not have a direct analytical form but could be solved via optimization or other non-linear solution techniques.